Cryptography

Data Encryption Standard (DES)

Cristhian Iza

School of Mathematical and Computational Sciences
Yachay Tech University

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Symmetric-Key

Stream Ciphers vs. Block Ciphers



UC 1. Cryptographic Foundations and Symmetric Cryptography

- Introduction to Cryptography and Security
- Classical Ciphers and Cryptanalysis
- Modular Arithmetic
- Shift and Affine Ciphers
- Symmetric Cryptography
- Stream Ciphers and basic attacks
- Block Ciphers and AES
- AES and Modes of Operation

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Bruce Schneier



Twofish,
Blowfish

Vincent Rijmen
and Joan Daemen



AES

Modern private
key encryption

Rivest, Shamir
& Aldeman



Public-k
encrypt

Ron
Rivest



Hashing

Whitfield
Diffie



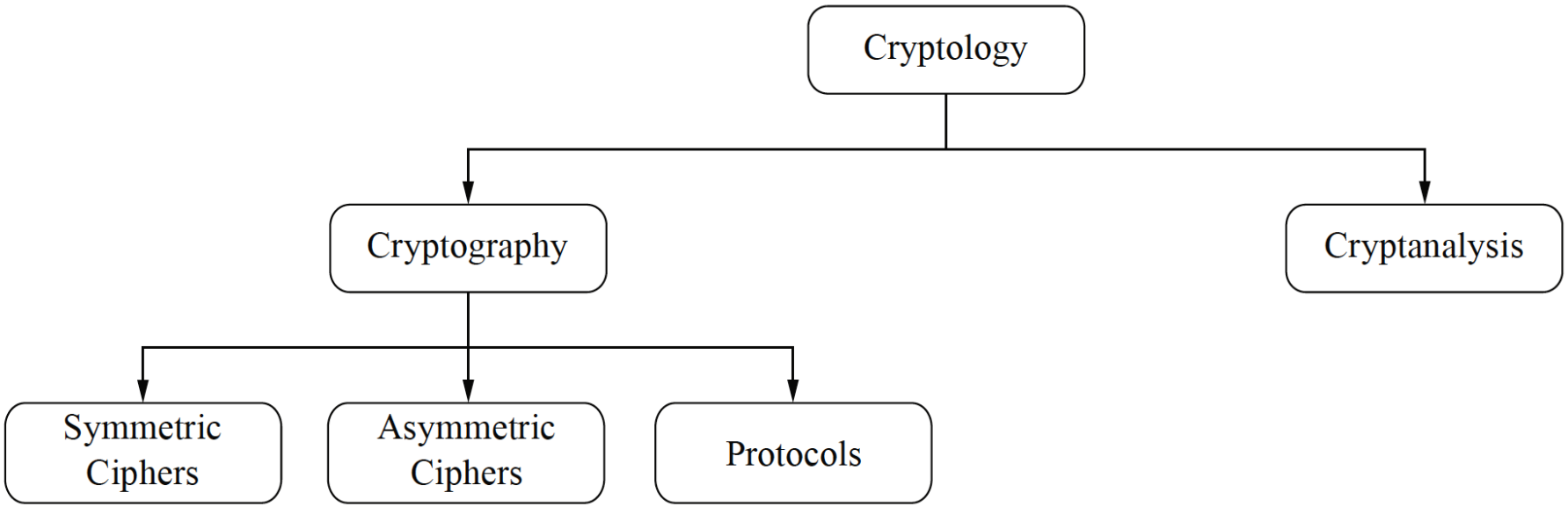
Key
interchange

Phil
Zimmerman



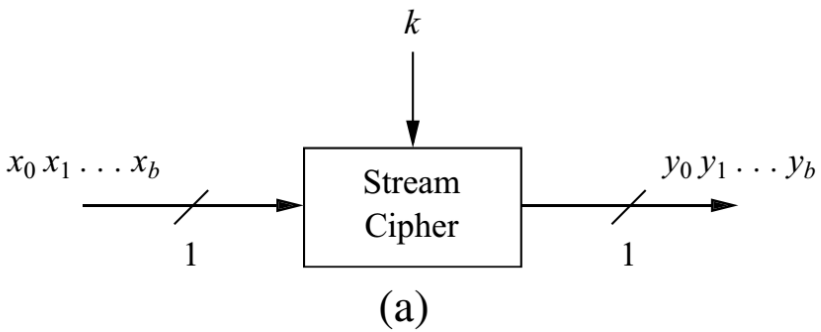
PGP
Encryption

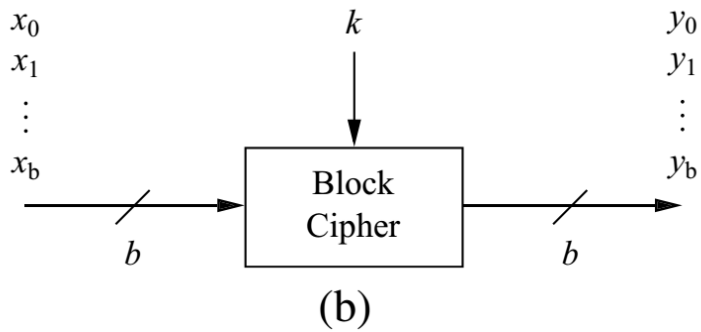
Stream Ciphers vs. Block Ciphers



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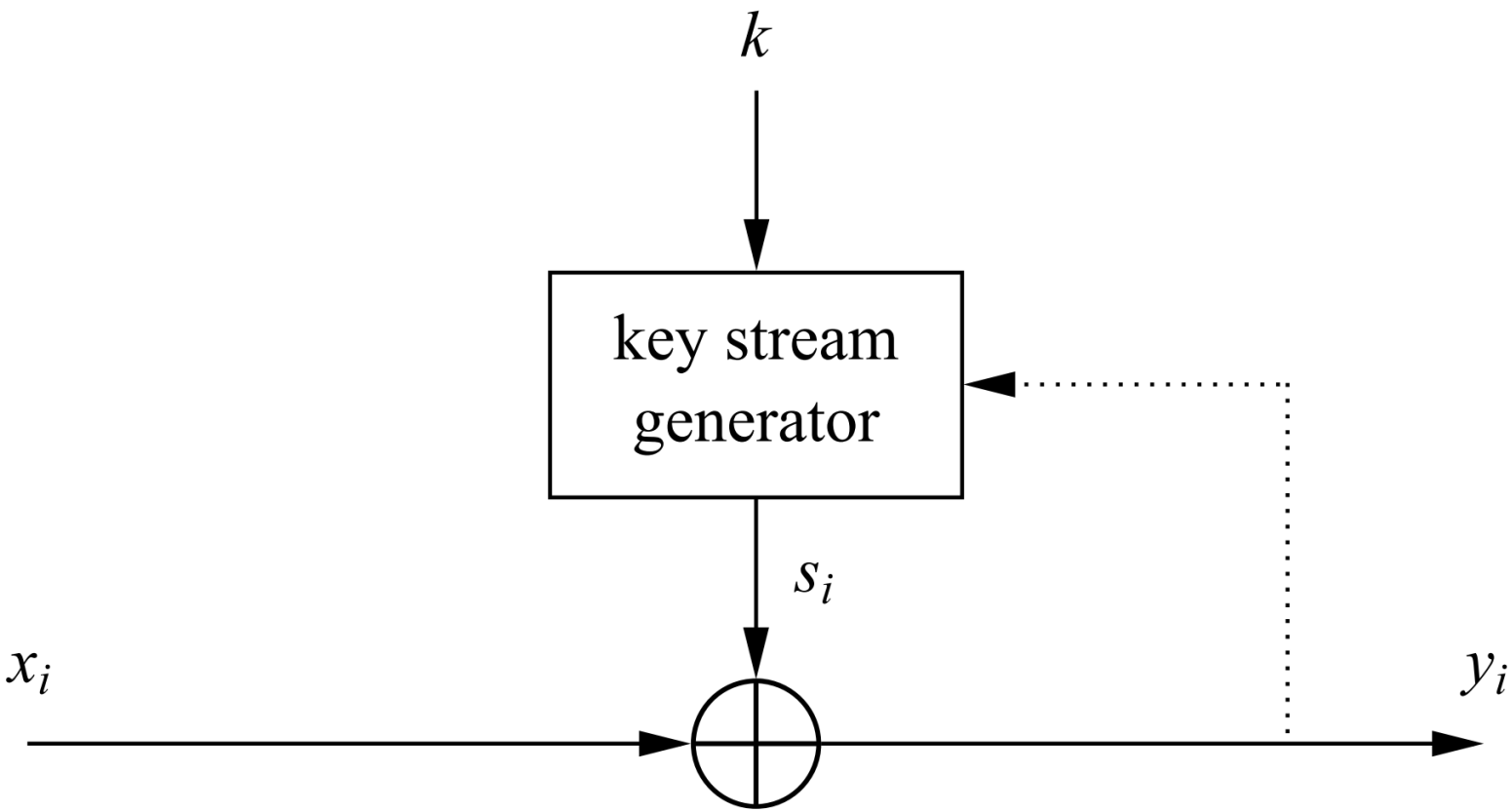
Stream Ciphers vs. Block Ciphers





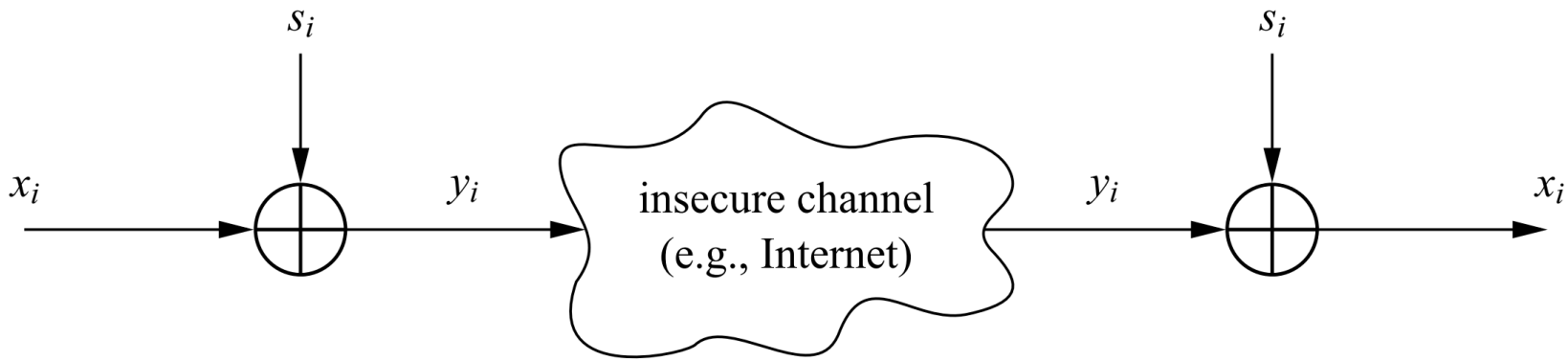
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Stream Ciphers vs. Block Ciphers



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Stream Ciphers vs. Block Ciphers



Encryption and Decryption with Stream Ciphers

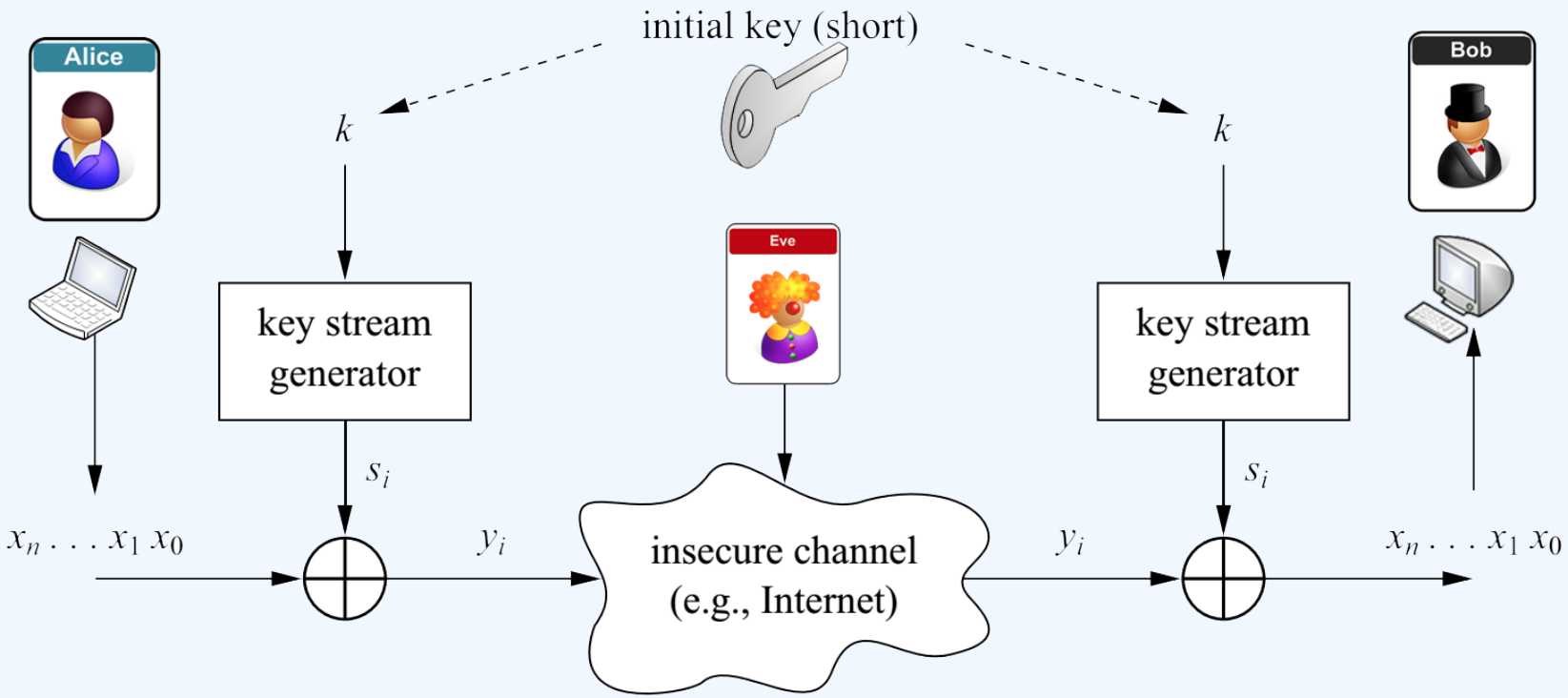
$i.e., x_i, y_i, s_i \in \{0, 1\}$

Encryption : $y_i = e_{s_i}(x_i) \equiv x_i + s_i \pmod 2$

Decryption : $x_i = d_{s_i}(y_i) \equiv y_i + s_i \pmod 2$

RNGs & Unbreakable Stream Cipher

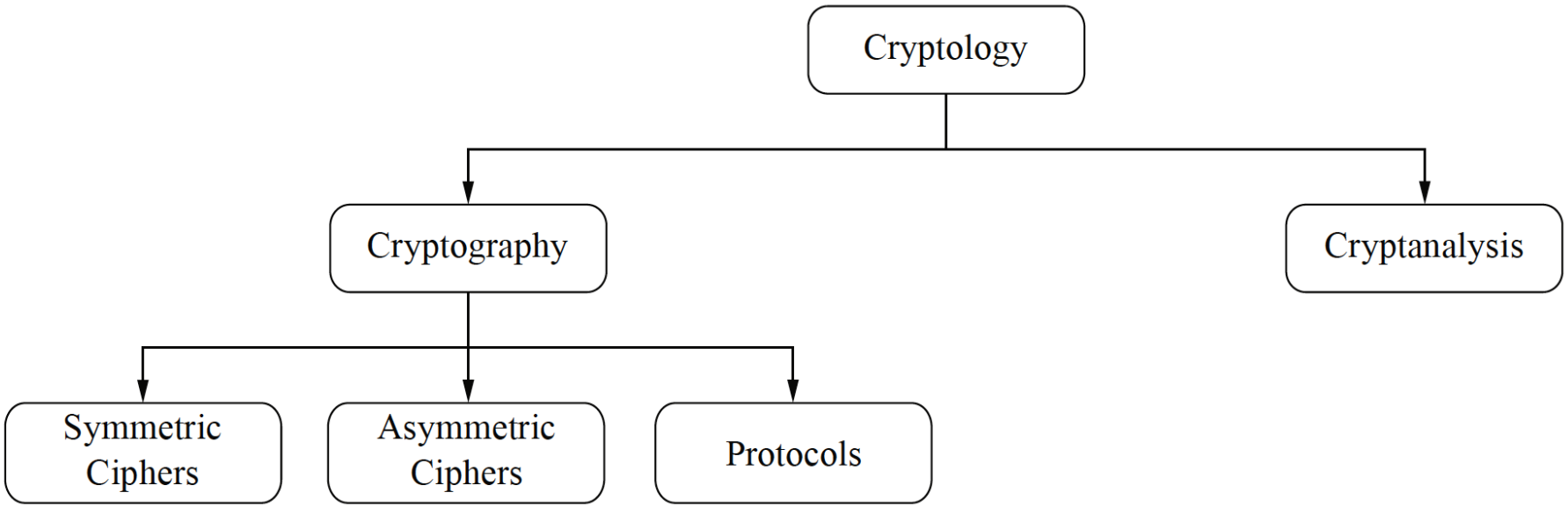
Practical Stream Ciphers



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Block Encryption Algorithms

Block Encryption Algorithm



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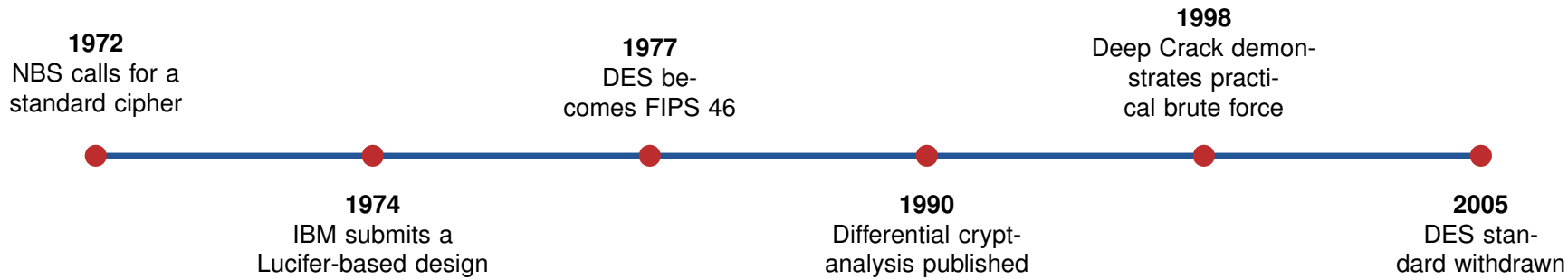
Block Encryption Algorithm

Symmetric encryption algorithms

- The most commonly used symmetric encryption algorithms are block ciphers.
- A block cipher processes the plaintext input in fixed-sized blocks and produces a block of ciphertext of equal size for each plaintext block.
- **Data Encryption Standard (DES)**, **triple DES (3DES)**, and **Advanced Encryption Standard (AES)**.

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Data Encryption Standard (DES)



DES grew out of IBM's Lucifer family and was standardized for public/commercial use during a major transition from secret government cryptography to open civilian cryptographic research.

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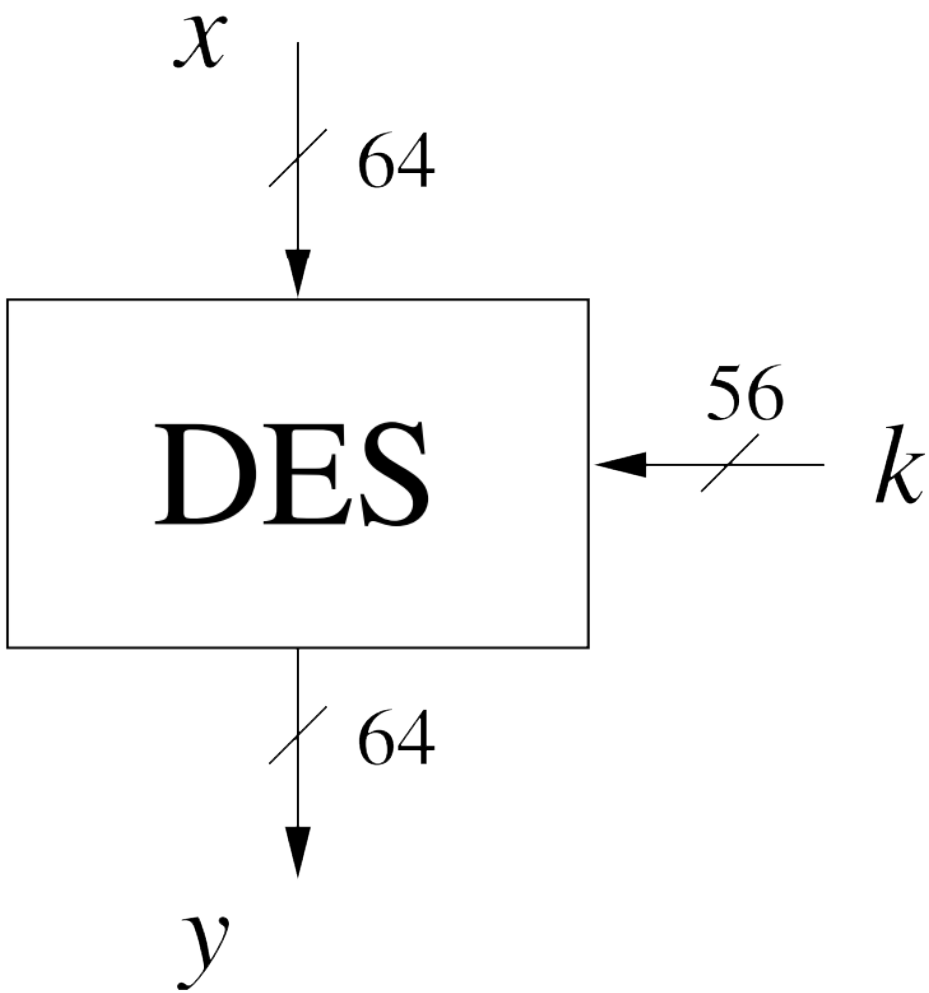
Symmetric Block Encryption Algorithms

Data Encryption Standard (DES)

- Unsecure today, but triple DES (3DES) is secure
- Studying DES helps us to understand many other symmetric algorithms.

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Symmetric Block Encryption Algorithms



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Symmetric Block Encryption Algorithms

Data Encryption Standard (DES)

- **How do we build a block cipher?**

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Symmetric Block Encryption Algorithms

Confusion

Confusion makes the relationship between the **key** and the **ciphertext** difficult to understand.

Small change in key $\rightarrow$ complex change in y

- Hides how the key affects the ciphertext.
- Makes key recovery more difficult.
- Mainly achieved by **substitution**.

Diffusion

Diffusion spreads the information of the **plaintext** across many parts of the ciphertext.

Change in one bit of x $\rightarrow$ changes in many y bits

- Hides patterns in the plaintext.
- Spreads the effect of each input bit.
- Mainly achieved by **permutation**.

Key idea

Confusion hides the relationship with the key, while diffusion hides patterns in the plaintext.

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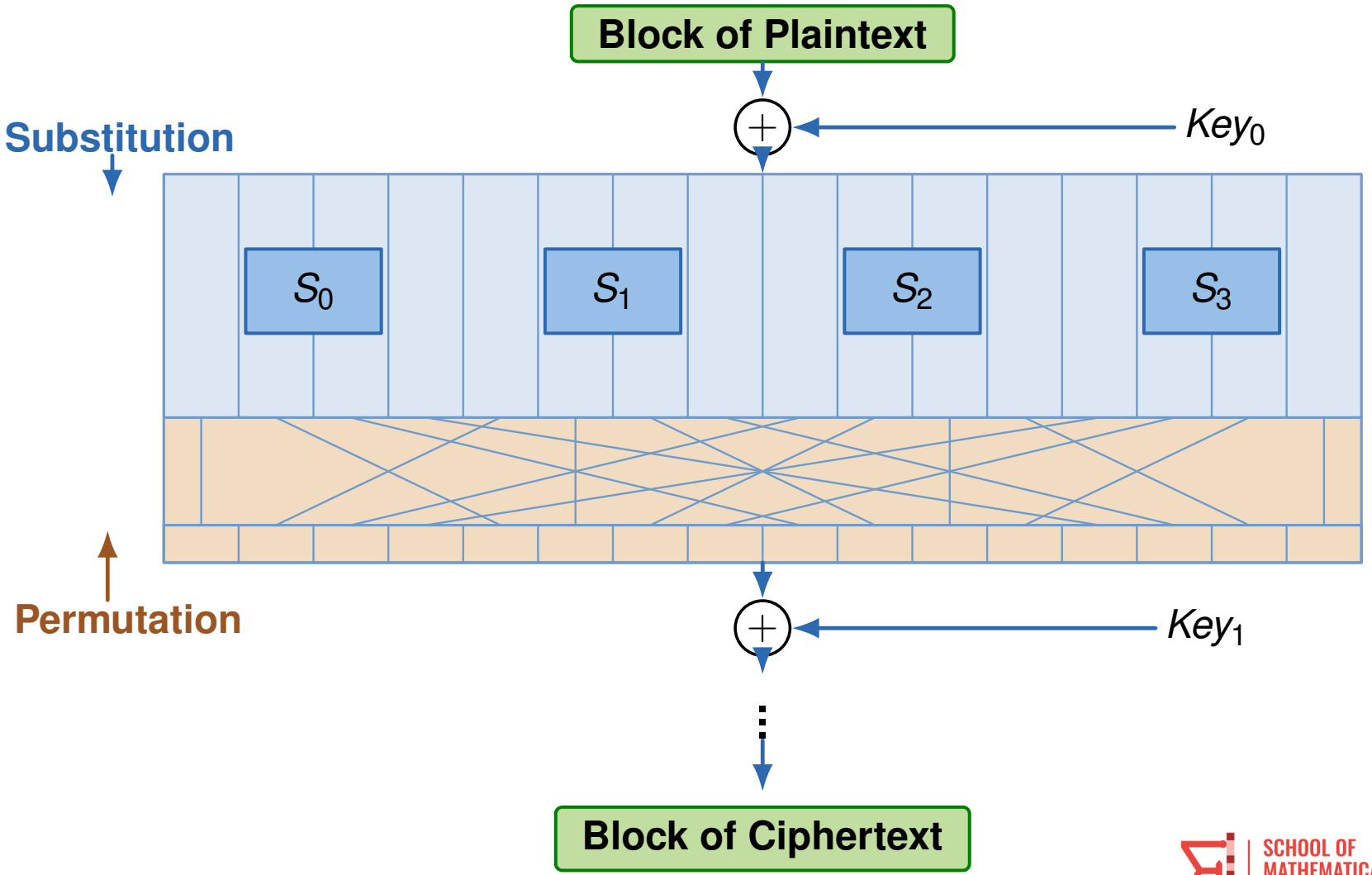
Symmetric Block Encryption Algorithms

Data Encryption Standard (DES)

- Confusion $\Rightarrow$ **Substitution**
- Diffusion $\Rightarrow$ **Permutation**

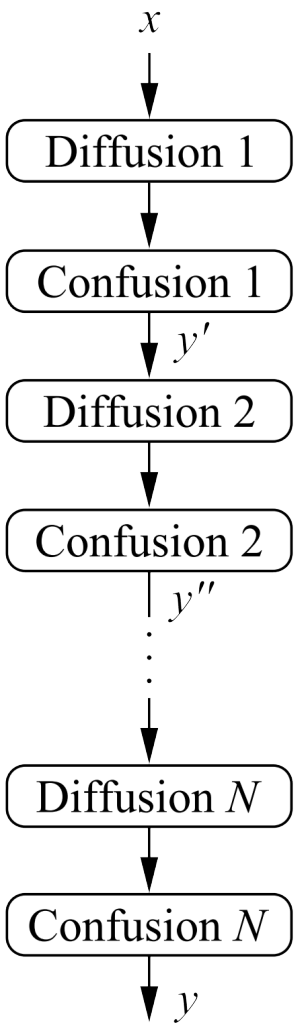
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Confusion and Difussion in a Block Cipher



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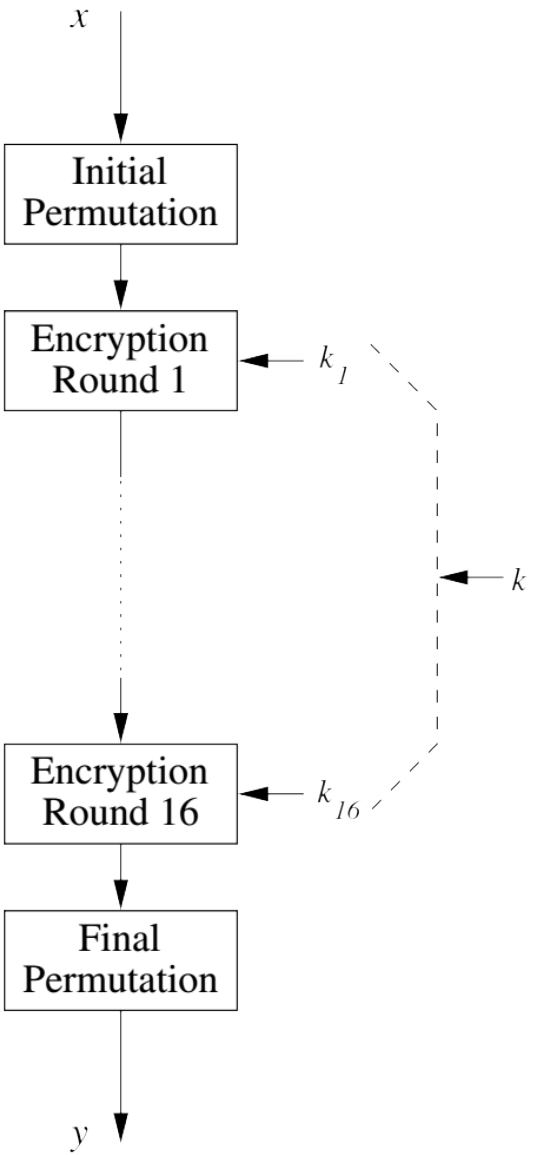
Principle of an N round product cipher



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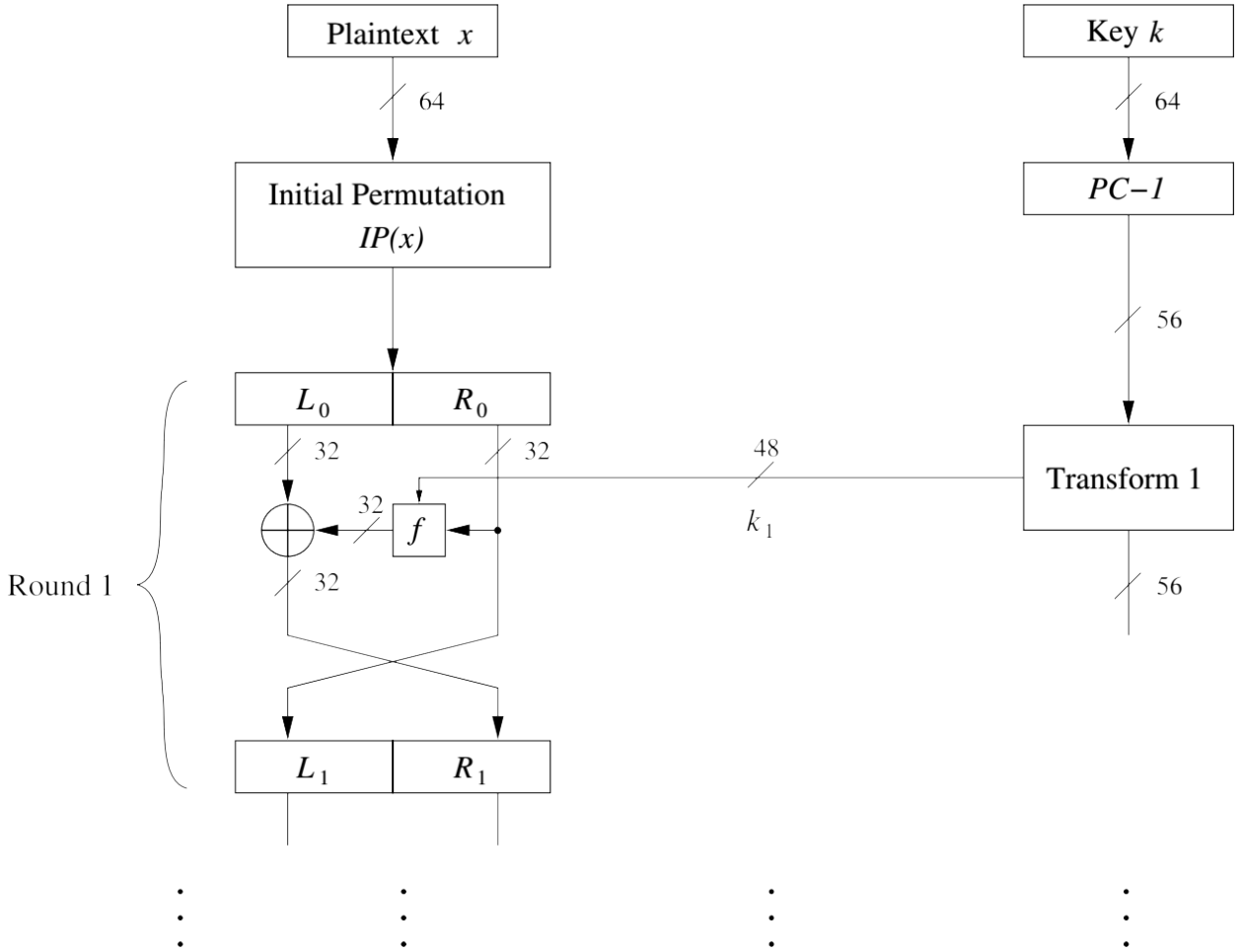
Data Encryption Standard (DES)

Overview of the DES Algorithm

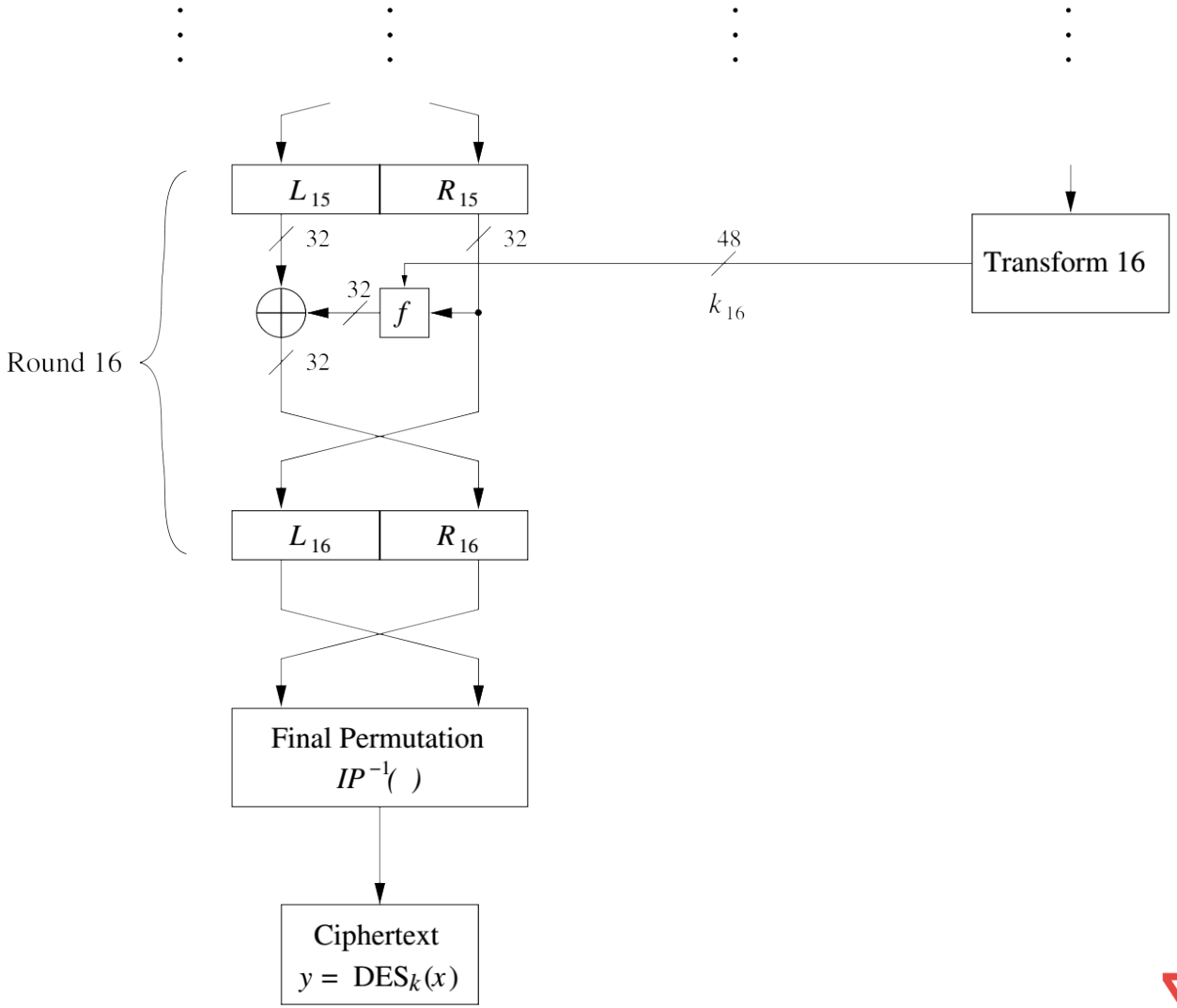


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Overview of the DES Algorithm



Overview of the DES Algorithm



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Symmetric block encryption algorithms

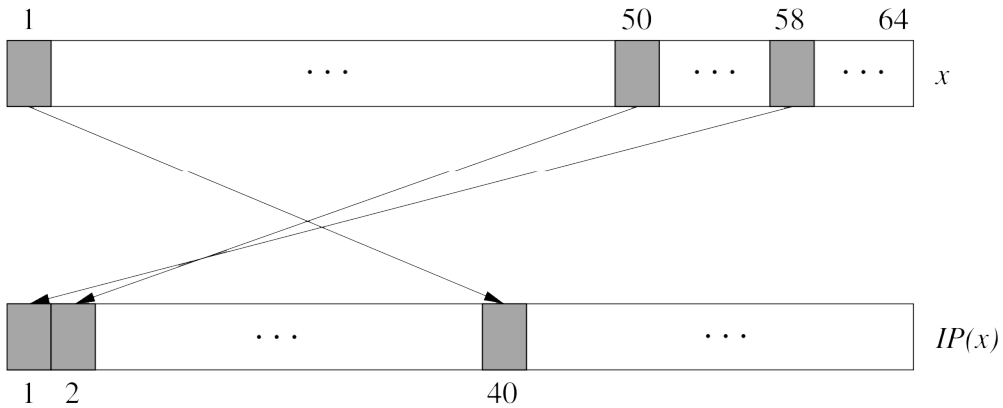
Internal Structure of Data Encryption Standard

- The plaintext is 64 bits in length.
- key is 64 bits (56 bits + 8 parity bits) in length.
- There are 16 rounds of processing.
- From the original 56-bit key, 16 subkeys are generated, one of which is used for each round.

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Initial Permutation of DES

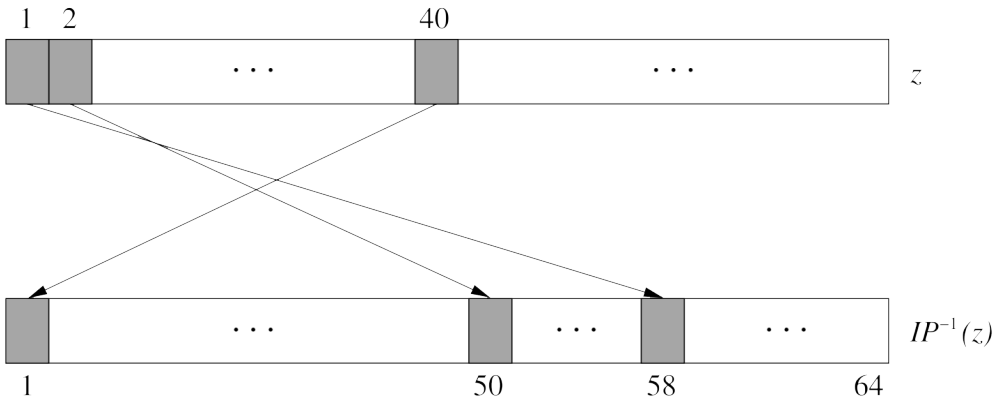
IP							
58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7



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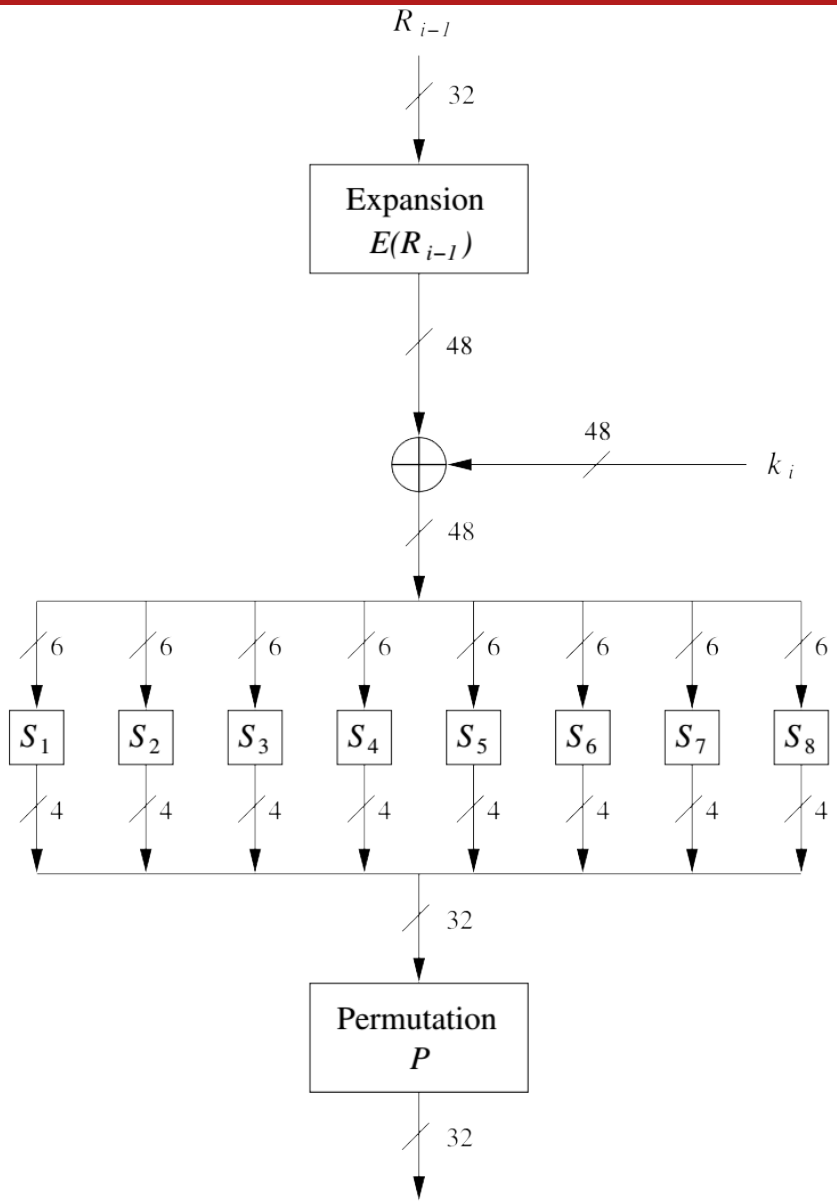
Final Permutation of DES

IP^{-1}							
40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25



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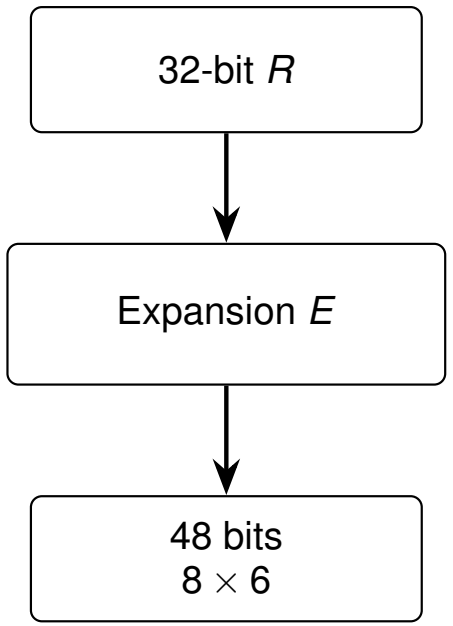
The *f*-Function of DES



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Step 1: Expansion $E(R_{i-1}$

- Input: 32 bits.
- Output: 48 bits.
- The input is divided into eight 4-bit regions.
- Boundary bits are repeated, forming eight overlapping 6-bit groups.
- Each 6-bit group becomes the input to one S-box.



fixed expansion with repeated boundary bits

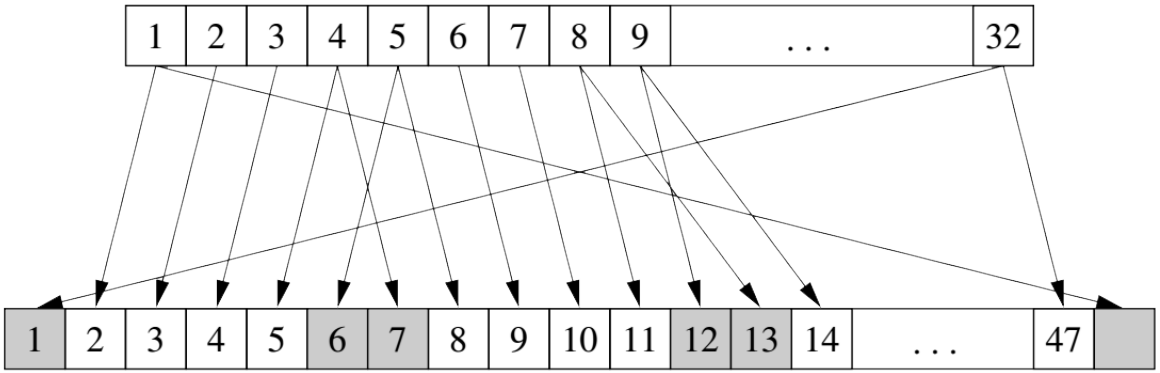
Purpose

Expand R from 32 to 48 bits for XOR with the 48-bit round key and allow boundary bits to affect two neighboring S-boxes.

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Step 1: Expansion $E(R_{i-1})$

E					
32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1



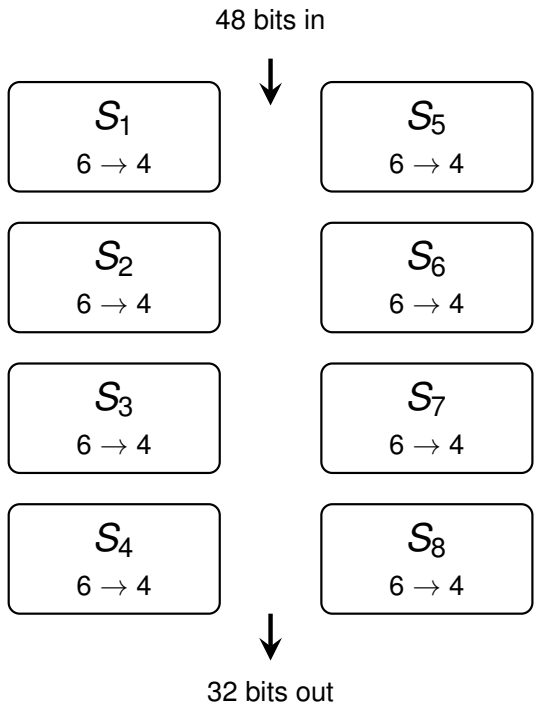
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Step 2: DES S-Boxes

- DES has **eight different S-boxes**.
- Each maps **6 input bits** to **4 output bits**.
- Eight S-boxes transform 48 bits into 32 bits.
- They are the **only nonlinear** elements of DES.

Why nonlinearity matters

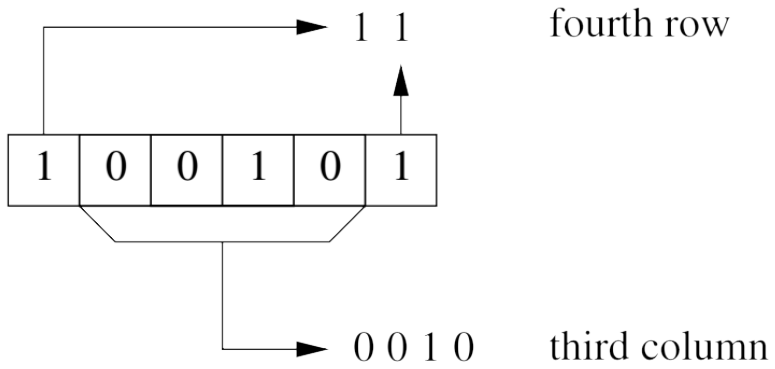
Without nonlinear components, the cipher could be modeled much more easily using linear equations.



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Step 2: DES S-Boxes

The S-box input $b = (100101)_2$ indicates the row $11_2 = 3$ and the column $0010_2 = 2$. If the input b is fed into S-box 1, the output is $S_1(37) = 8 = (1000)_2$.



S-box S_1

S_1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	01	10	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13

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Step 2: DES S-Boxes

S-box S_1

S_1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	01	10	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13

S-box S_2

S_2	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	15	01	08	14	06	11	03	04	09	07	02	13	12	00	05	10
1	03	13	04	07	15	02	08	14	12	00	01	10	06	09	11	05
2	00	14	07	11	10	04	13	01	05	08	12	06	09	03	02	15
3	13	08	10	01	03	15	04	02	11	06	07	12	00	05	14	09

S-box S_3

S_3	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	10	00	09	14	06	03	15	05	01	13	12	07	11	04	02	08
1	13	07	00	09	03	04	06	10	02	08	05	14	12	11	15	01
2	13	06	04	09	08	15	03	00	11	01	02	12	05	10	14	07
3	01	10	13	00	06	09	08	07	04	15	14	03	11	05	02	12

S-box S_4

S_4	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	07	13	14	03	00	06	09	10	01	02	08	05	11	12	04	15
1	13	08	11	05	06	15	00	03	04	07	02	12	01	10	14	09
2	10	06	09	00	12	11	07	13	15	01	03	14	05	02	08	04
3	03	15	00	06	10	01	13	08	09	04	05	11	12	07	02	14

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Step 2: DES S-Boxes

S-box S_5

S_5	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	02	12	04	01	07	10	11	06	08	05	03	15	13	00	14	09
1	14	11	02	12	04	07	13	01	05	00	15	10	03	09	08	06
2	04	02	01	11	10	13	07	08	15	09	12	05	06	03	00	14
3	11	08	12	07	01	14	02	13	06	15	00	09	10	04	05	03

S-box S_6

S_6	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	12	01	10	15	09	02	06	08	00	13	03	04	14	07	05	11
1	10	15	04	02	07	12	09	05	06	01	13	14	00	11	03	08
2	09	14	15	05	02	08	12	03	07	00	04	10	01	13	11	06
3	04	03	02	12	09	05	15	10	11	14	01	07	06	00	08	13

S-box S_7

S_7	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	04	11	02	14	15	00	08	13	03	12	09	07	05	10	06	01
1	13	00	11	07	04	09	01	10	14	03	05	12	02	15	08	06
2	01	04	11	13	12	03	07	14	10	15	06	08	00	05	09	02
3	06	11	13	08	01	04	10	07	09	05	00	15	14	02	03	12

S-box S_8

S_8	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	13	02	08	04	06	15	11	01	10	09	03	14	05	00	12	07
1	01	15	13	08	10	03	07	04	12	05	06	11	00	14	09	02
2	07	11	04	01	09	12	14	02	00	06	10	13	15	03	05	08
3	02	01	14	07	04	10	08	13	15	12	09	00	03	05	06	11

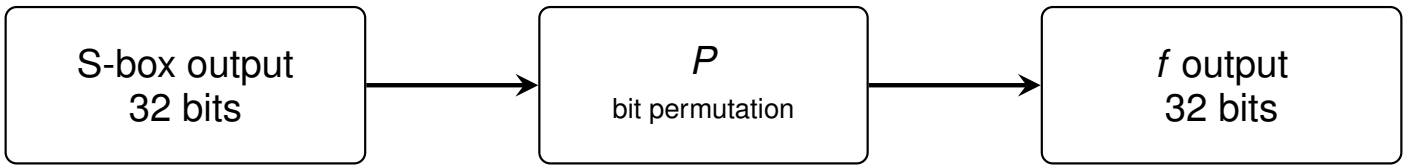
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Step 3: Permutation P

- Input: 32 S-box output bits.
- Output: the same 32 bits, rearranged.
- The output bits of each S-box are distributed so that they influence different S-boxes in the next round.

Role of P

Unlike IP and IP^{-1}, the internal permutation P contributes directly to **diffusion**.



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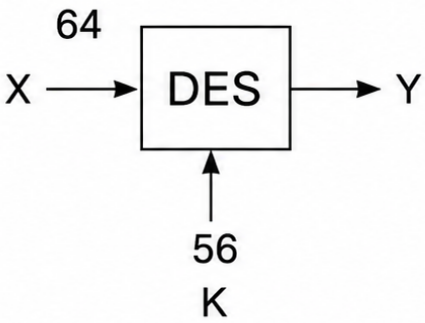
Step 3: Permutation P

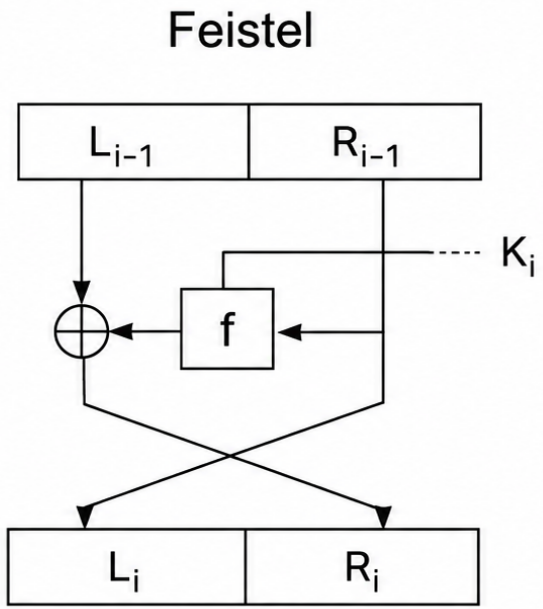
Finally, the `32-bit` output is permuted bitwise according to the P permutation, which is given in Table

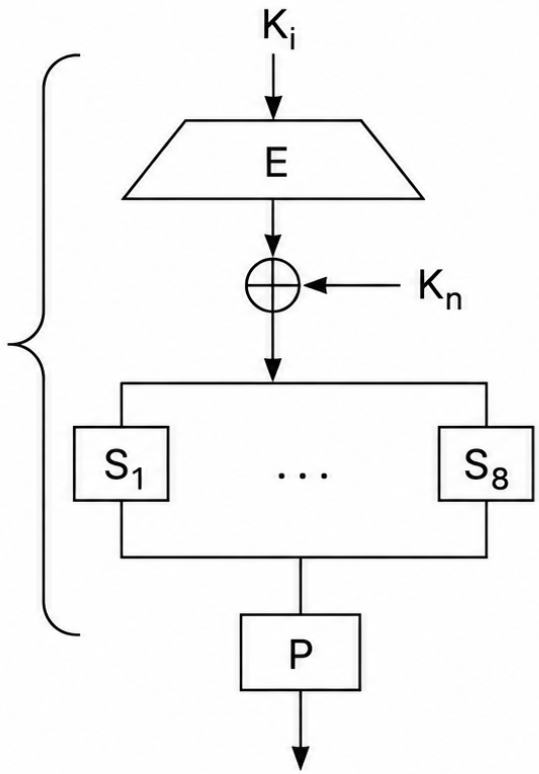
P							
16	7	20	21	29	12	28	17
1	15	23	26	5	18	31	10
2	8	24	14	32	27	3	9
19	13	30	6	22	11	4	25

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DES Key Schedule

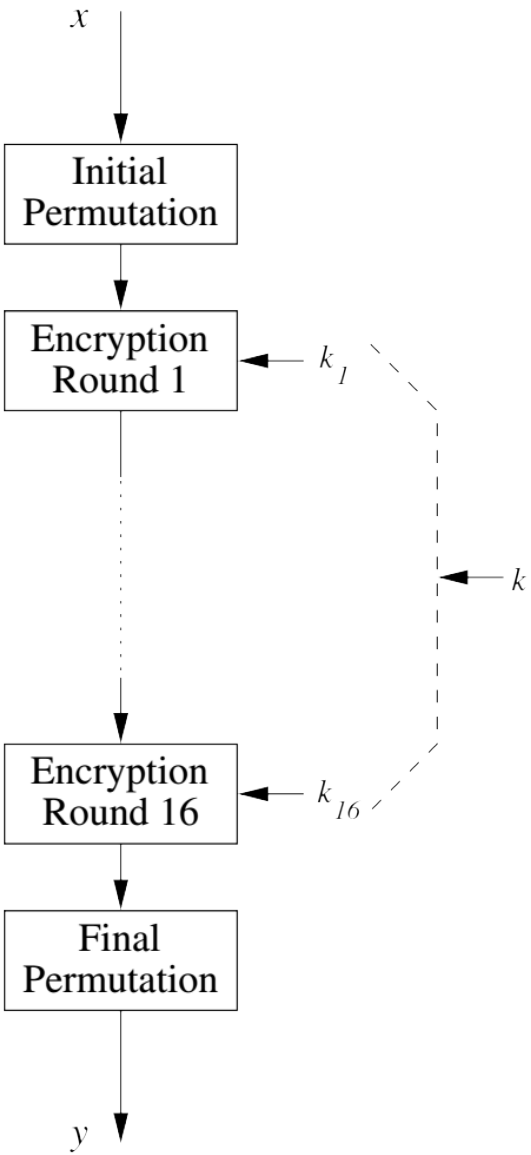






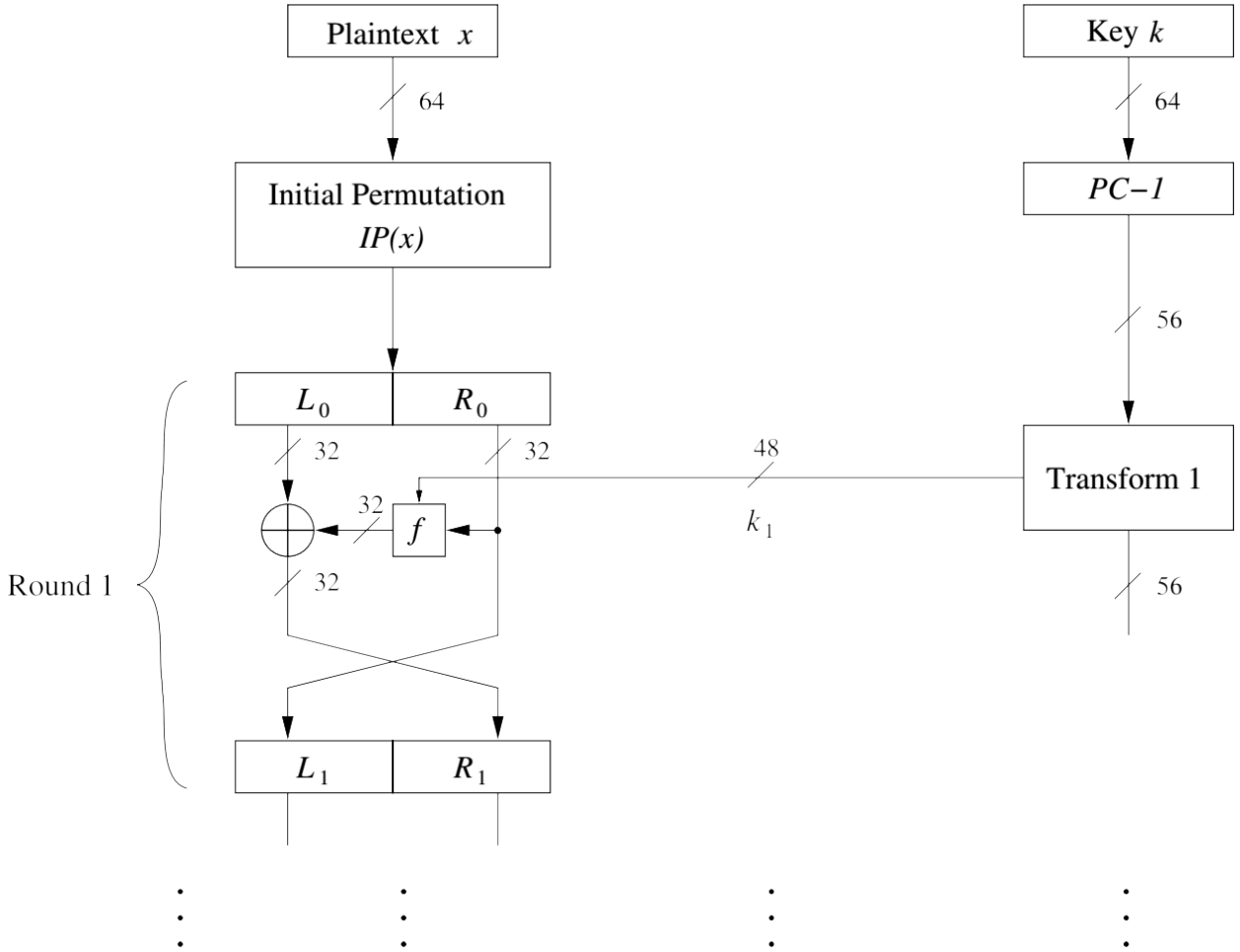
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DES Key ScheduleS



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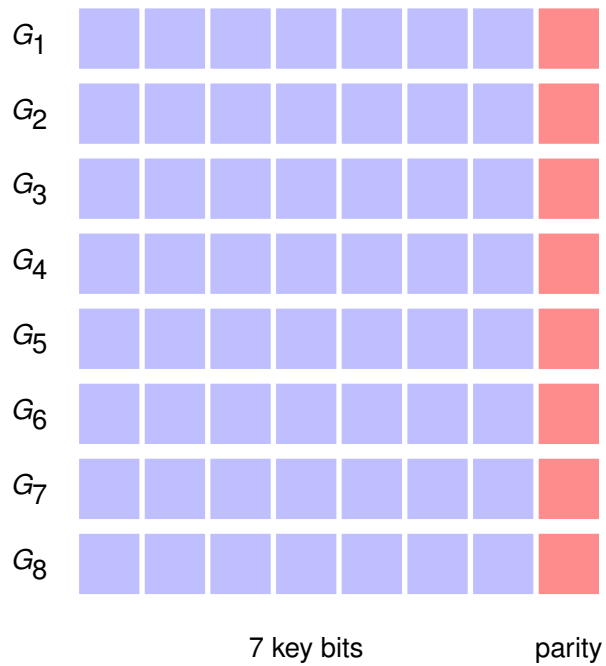
The *key*-Schedule of DES



DES Key Schedule

- DES keys are represented using **64 bits**.
- Every eighth bit is used as a **parity bit**.
- The 8 parity bits do **not** contribute to the cryptographic key space.
- Therefore, the effective key length is

$$64 - 8 = \boxed{56 \text{ bits}}.$$



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DES Key Schedule

The *key*-Schedule of DES

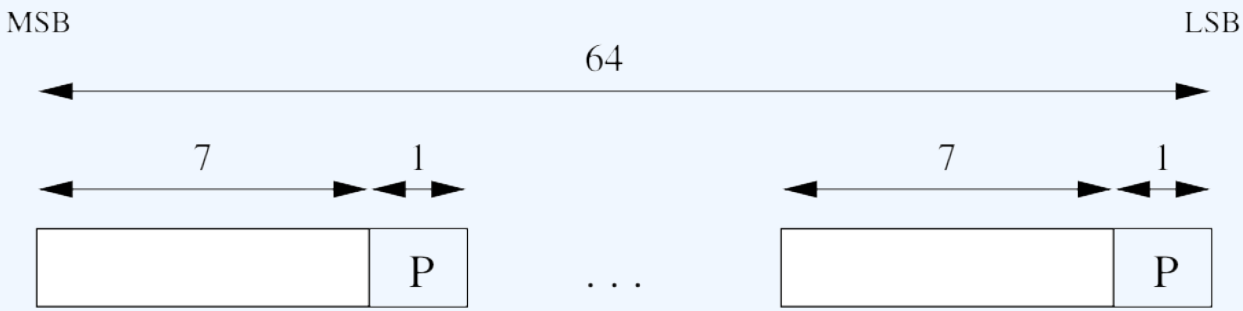
- How to compute the 16 subkeys $k_1, \ldots, k_{16}$?
- Key schedule consists of simple operations

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DES Key Schedule

PC – 1 Permuted Choice 1

- Drops bits $8, 16, \ldots, 64$
- Effective key length of DES: $64 - 8 = 56$



P = parity bit

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DES Key Schedule

Initial key permutation $PC-1$ of DES

- The 64-bit key is first reduced to 56 bits, the parity bits are stripped in the initial $PC-1$ permutation.
- The name $PC-1$ stands for permuted choice one.

$PC-1$							
57	49	41	33	25	17	9	1
58	50	42	34	26	18	10	2
59	51	43	35	27	19	11	3
60	52	44	36	63	55	47	39
31	23	15	7	62	54	46	38
30	22	14	6	61	53	45	37
29	21	13	5	28	20	12	4

DES Key Schedule

LS_i

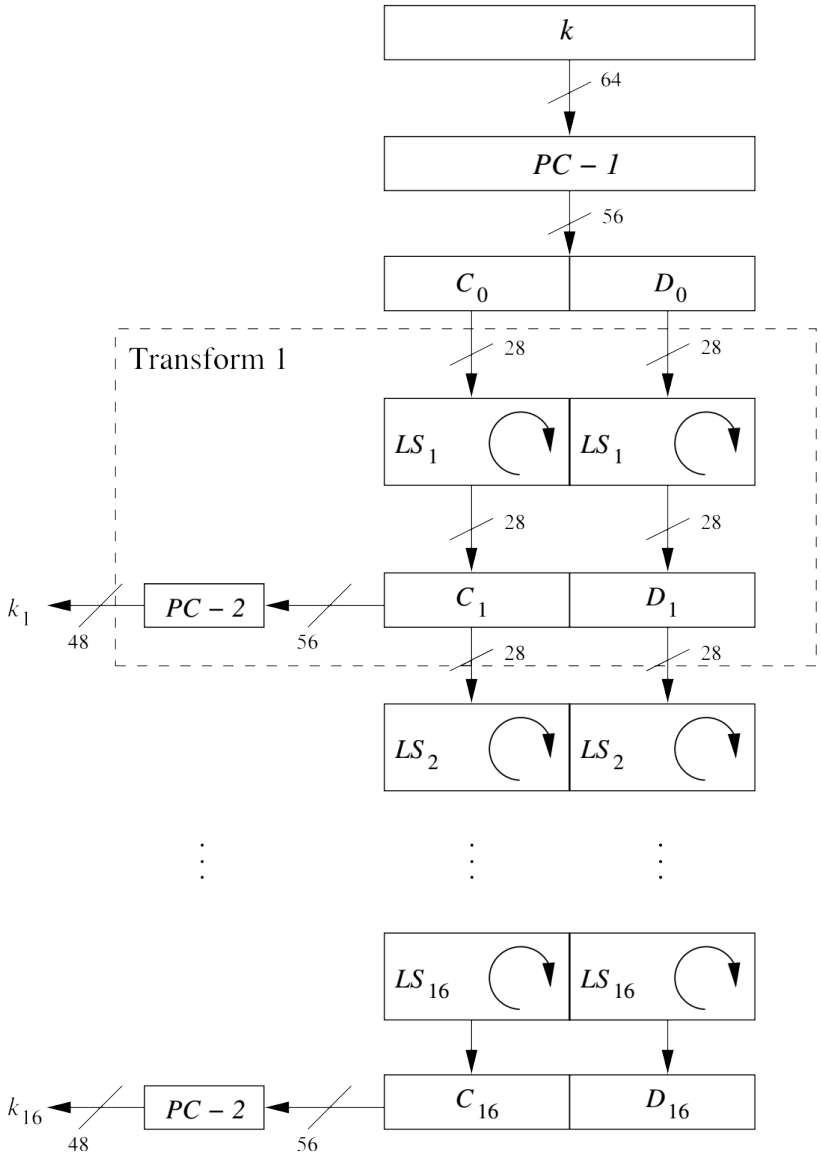
- Left shift (actually: left rotation)

$$LS_i = \begin{cases} 1, & \text{if } i \in \{1, 2, 9, 16\}, \\ 2, & \text{otherwise.} \end{cases} \tag{1}$$

- Total number of bit positions shifted: $4 \cdot 1 + 12 \cdot 2 = 28$

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DES Key Schedule



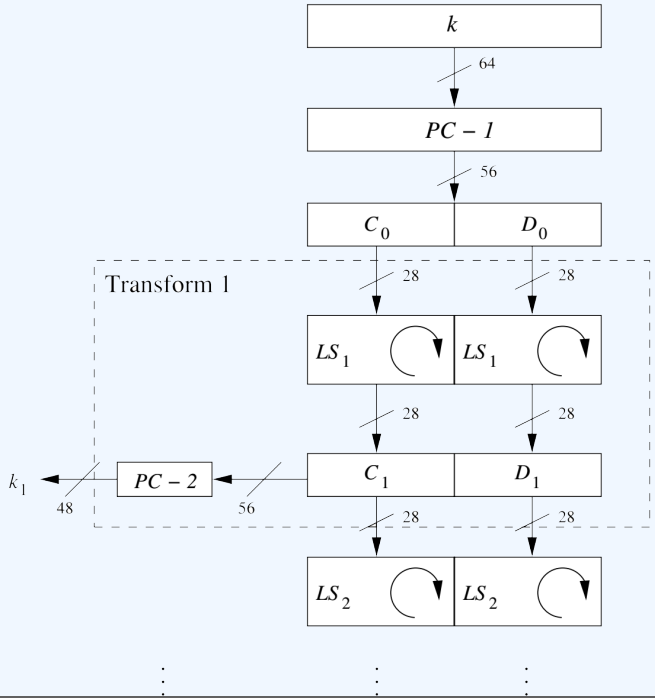
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DES Key Schedule

Key schedule for DES

The two 28-bit halves are cyclically shifted, i.e., rotated, left by one or two bit positions depending on the round i:

- Rounds $i = 1, 2, 9, 16$, the two halves are rotated left by one bit.
- In others rounds $i \neq 1, 2, 9, 16$, the two halves are rotated left by two bits.
- Total rotation positions is $4 \cdot 1 + 12 \cdot 2 = 28 \rightarrow C_0 = C_{16}$ and $D_0 = D_{16}$.



DES Key Schedule

Round key permutation $PC-2$ of DES

- To derive the 48-bit round keys k_i, the two halves are permuted bitwise again with $PC-2$, which stands for permuted choice 2.
- $PC-2$ permutes the 56 input bits coming from C_i and D_i and ignores 8 of them.

$PC-2$							
14	17	11	24	1	5	3	28
15	6	21	10	23	19	12	4
26	8	16	7	27	20	13	2
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

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DES Key Schedule

Key schedule for DES

Each key $k_1, \ldots, k_{16}$ is merely a permutation of the original 56 bit key.

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DES Decryption

Key Idea

DES uses the **same structure** for encryption and decryption. The only change is the **order of the round keys**.

- The ciphertext is used as the input.
- The same 16 Feistel rounds are applied.
- The round keys are used in **reverse order**.

Round-Key Order

Encryption: $K_1, K_2, \ldots, K_{15}, K_{16}$

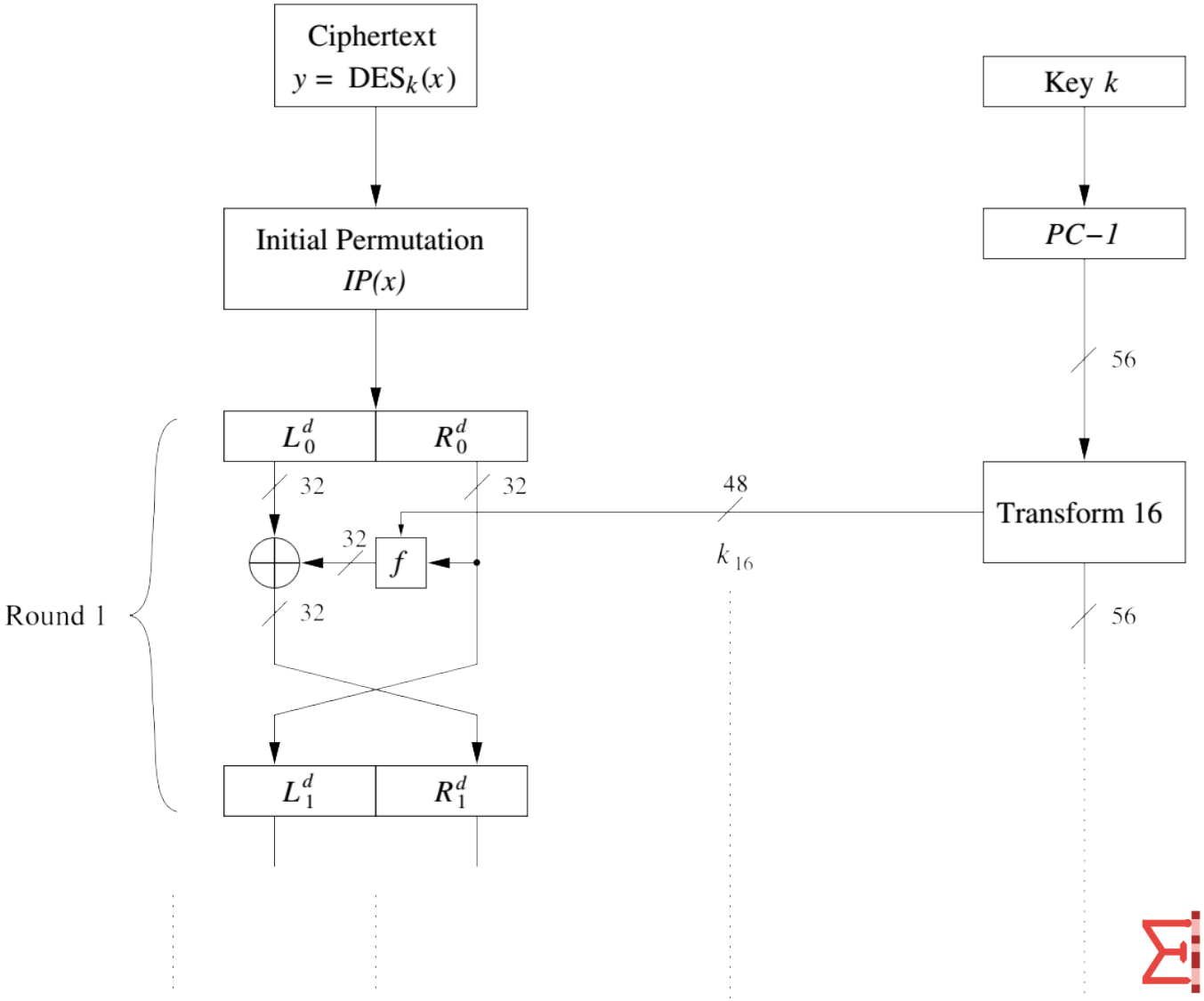
Decryption: $K_{16}, K_{15}, \ldots, K_2, K_1$

Feistel Property

No inverse of the round function f is required. Reversing the subkey order is sufficient to recover the plaintext.

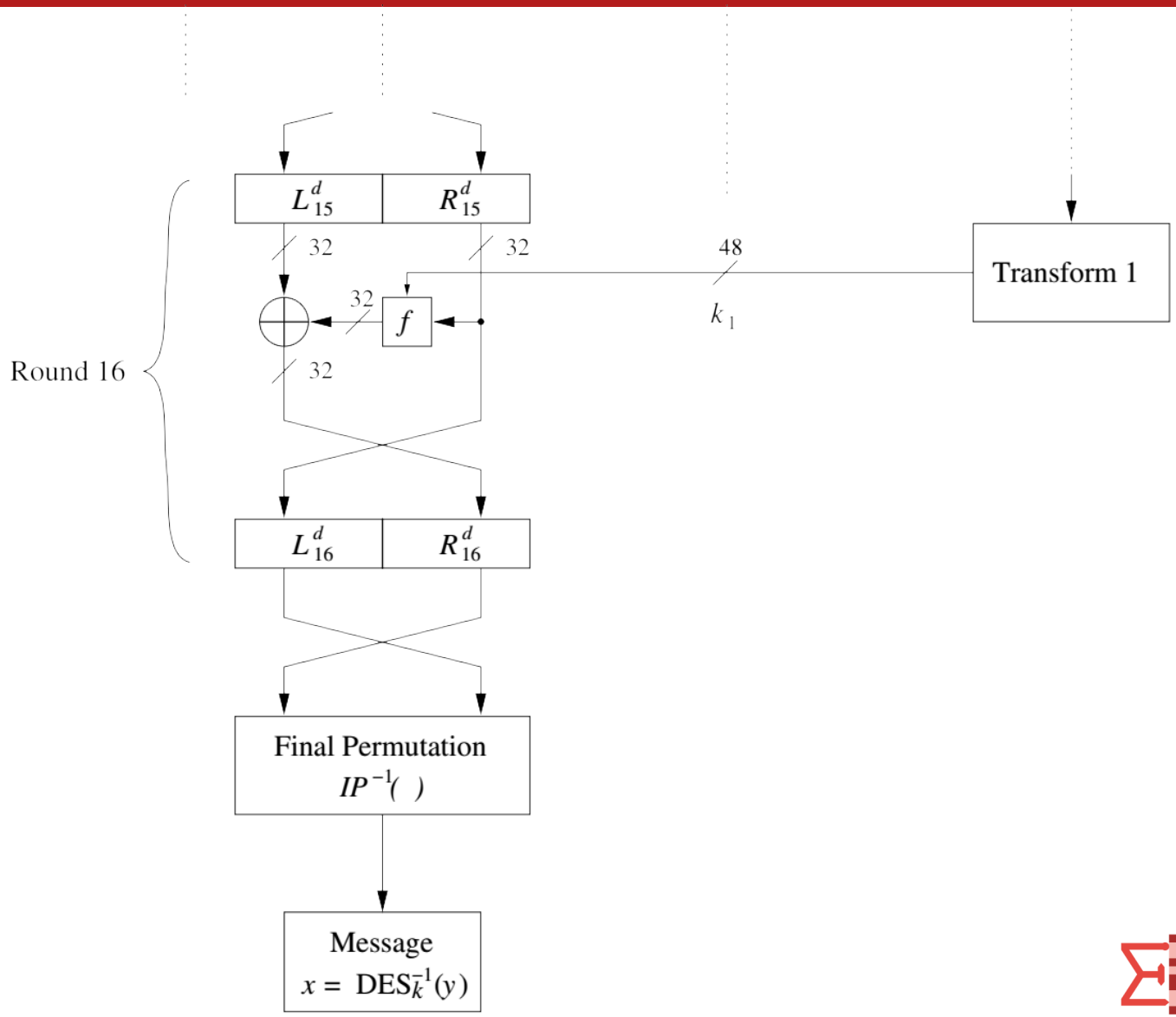
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Decryption of DES



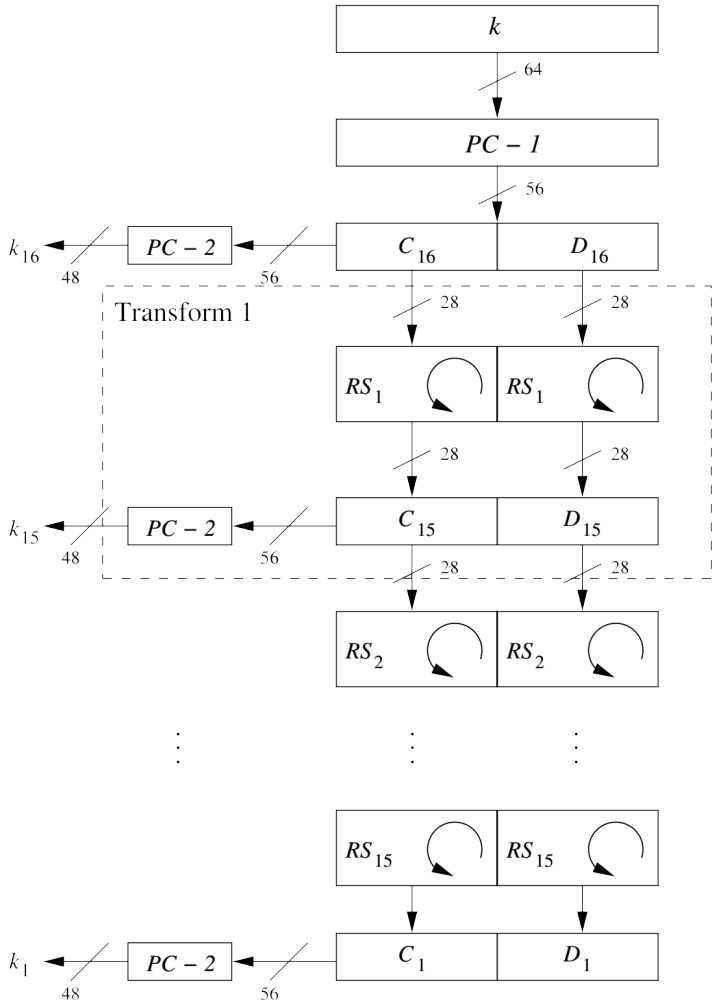
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Decryption of DES



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Reversed key schedule for decryption of DES



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Strength and Weaknesses of DES

Algorithm Strength

- DES was designed to resist several forms of cryptanalysis.
- Its S-boxes provide strong **nonlinearity**.
- Differential and linear cryptanalysis are possible, but require a very large amount of data.

Main Weakness

- DES has only a **56-bit effective key**.
- The key space contains

$$2^{56}$$

possible keys.

- Modern hardware can perform an exhaustive **brute-force search**.

Key Point

The main practical weakness of DES is not its Feistel structure, but its **short key length**.

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DES and Exhaustive Key Search

Brute-Force Attack

- DES has an effective key length of only **56 bits**.
- An exhaustive search may require testing up to

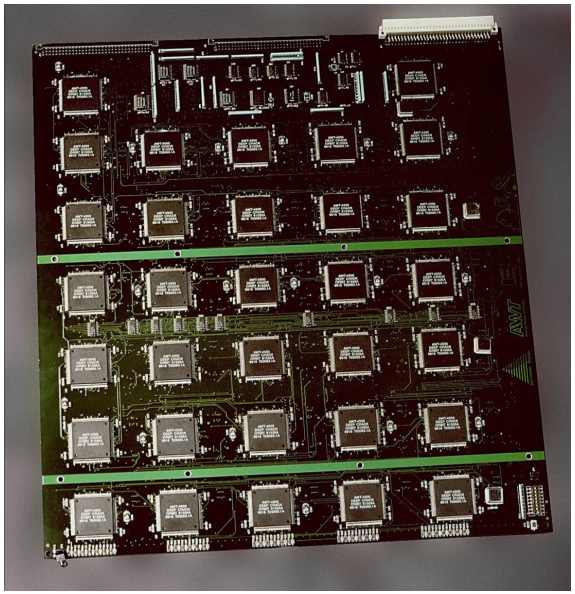
$$2^{56} \approx 7.2 \times 10^{16}$$

possible keys.

- The attack does not exploit a weakness in the DES structure; it simply tries keys until the correct one is found.

1998: DES Was Broken in Practice

The Electronic Frontier Foundation (EFF) built a dedicated DES-cracking machine for about **US$250,000**.



EFF DES Cracker
"Deep Crack" – 1998

DES key recovered in 56 hours

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DES Brute Force with COPACOBANA

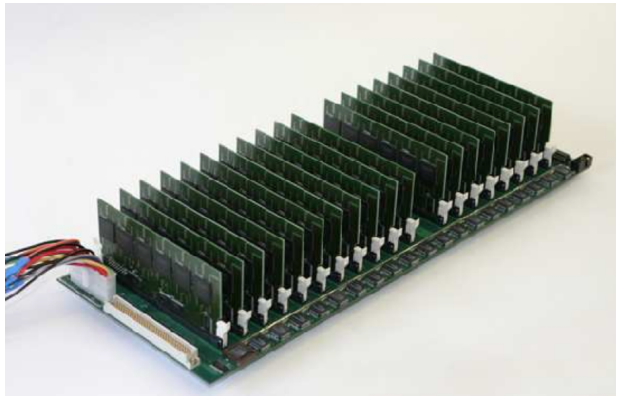
COPACOBANA

- **CO**st-Optimized **PA**rallel **CO**de **B**reaker.
- Introduced in 2006 as a low-cost, FPGA-based cryptanalysis machine.
- Used massive parallelism to test many DES keys simultaneously.
- Estimated hardware cost:

  about US$10,000

- Average DES key-search time:

  $\approx$ 6.4 days



COPACOBANA
FPGA-based code breaker, 2006

Key Lesson

Parallel hardware made exhaustive search increasingly affordable, confirming that a **56-bit DES key is too short**.

From DES to Modern Encryption (cont.)

Alternatives to DES

- **Triple DES (3DES)**
  - Applies DES three times to increase security.
  - Developed as a transitional replacement for DES.
  - Now considered a **legacy algorithm** and deprecated for new systems.
- **Advanced Encryption Standard (AES)**
  - Designed as the modern replacement for DES.
  - Supports **128-, 192-, and 256-bit keys**.
  - Widely used as the standard block cipher today.

Evolution

DES $\longrightarrow$ 3DES $\longrightarrow$ AES

56 bits legacy modern standard

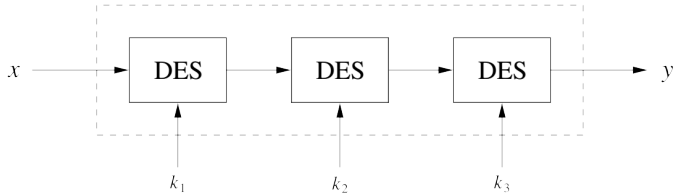
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Triple DES (3DES) (cont.)

Why Triple DES?

- 3DES was introduced to extend the lifetime of DES without designing a completely new cipher.
- It applies the DES algorithm **three times** to each 64-bit block.
- The common construction is **Encrypt–Decrypt–Encrypt (EDE)**:

$$C = E_{K_3}\left(D_{K_2}\left(E_{K_1}(P)\right)\right)$$



Triple-DES EDE structure

Security

3DES provides much stronger protection than DES against brute-force key search, but it is now considered a **legacy cipher**.

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Triple DES (3DES) (cont.)

Historical Role

DES $\longrightarrow$ **3DES** $\longrightarrow$ **AES**

3DES served as a transitional solution before AES became the modern standard.

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Lab 1: DES Implementation (cont.)

Objective

Implement the **Data Encryption Standard (DES)** from its specification and understand how its main components work together.

Main Tasks

- Implement the DES key schedule.
- Implement the 16 Feistel rounds.
- Implement:
  - Initial permutation IP
  - Expansion E
  - S-boxes $S_1, \ldots, S_8$
  - Permutation P
  - Final permutation IP^{-1}
- Implement encryption and decryption.

Implementation Rules

- DES must be programmed **from the specification**.
- Do not use existing DES implementations or cryptographic libraries for the DES core.
- Validate the implementation using known test vectors.

The following items must be available in the student's GitHub repository:

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Lab 1: DES Implementation (cont.)

Deliverables

1. Source code of the DES library.
2. Automatic test suite.
3. `README.md` with installation and usage instructions.
4. Laboratory report in PDF format.

The repository URL must be submitted through the course platform and included in the laboratory report.

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Modes of Operation & Brute-Force

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From Lab 1 to Lab 2

Lab 1: Build DES

- Implement DES from the specification.
- Validate 64-bit block encryption/decryption.
- Understand the key schedule and Feistel rounds.

Lab 2: Use and Attack DES

- Encrypt messages longer than one block.
- Implement **ECB** and **CBC**.
- Perform a controlled **brute-force attack**.
- Evaluate CPU parallelism experimentally.

Key idea

Reuse the DES library developed in Lab 1. Do not implement DES again.

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From One DES Block to a Complete Message

- DES processes exactly one **64-bit block**:

$$64 \text{ bits} = 8 \text{ bytes}.$$

- Real messages contain an arbitrary number of bytes.
- A **mode of operation** defines how multiple DES blocks are processed.
- The final partial block must be completed before encryption.

PKCS#7 Padding

If p bytes are required, append p copies of the byte value p.

Example:

```
41 42 43 44 45
```

becomes

```
41 42 43 44 45 03 03 03.
```

Implementation

Implement `pad()` and `unpad()`; invalid padding must be rejected.

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PKCS#7: Padding an Exact 8-Byte Message

Original data block

`41 42 43 44 45 46 47 48`

8 bytes

+

Complete padding block

`08 08 08 08 08 08 08 08`

8 bytes of padding

Padded message: 16 **bytes** = 2 **DES blocks**

Why not add zero bytes?

If no extra block were added, a valid message ending in `01` (e.g., `. . . 47 01`) could be confused with "one byte of padding." PKCS#7 avoids this ambiguity by **always adding padding**.

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Electronic Codebook Mode (ECB)

Encryption

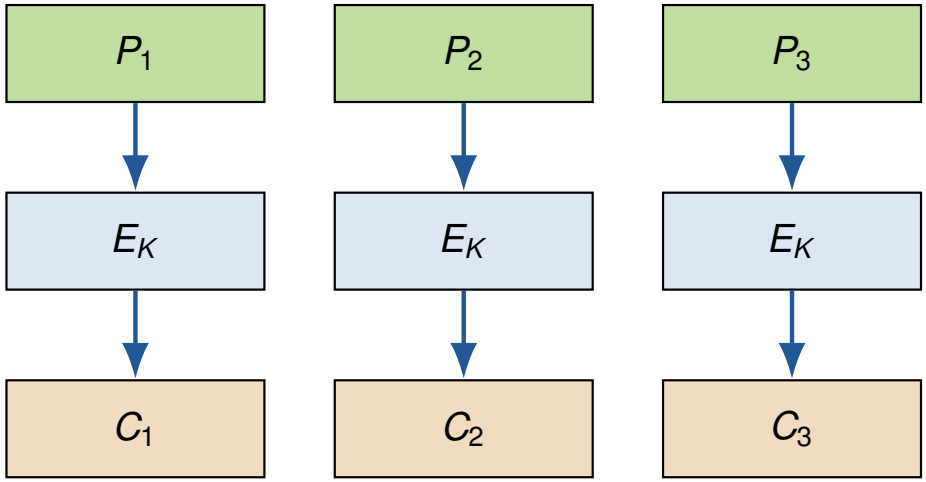
$$C_i = E_K(P_i)$$

Each plaintext block is encrypted **independently**.

Main weakness

$$P_i = P_j \implies C_i = C_j$$

Repeated plaintext blocks produce repeated ciphertext blocks.



No chaining between blocks

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Cipher Block Chaining Mode (CBC)

Encryption

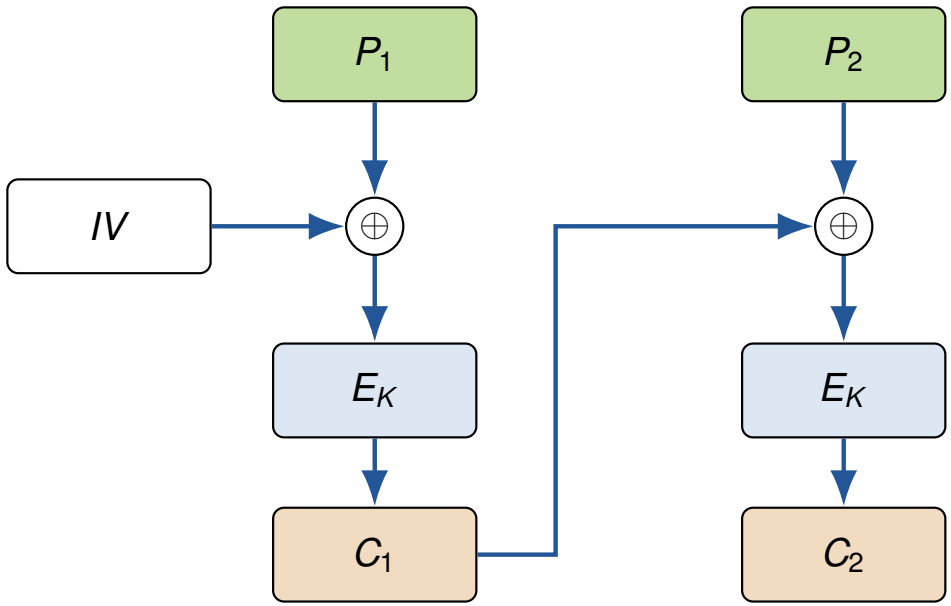
$$C_0 = IV, \qquad C_i = E_K(P_i \oplus C_{i-1})$$

Decryption

$$P_i = D_K(C_i) \oplus C_{i-1}$$

Initialization Vector

A different IV changes the first encrypted block even when the key and plaintext are unchanged.



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Experiments: What Must You Observe?

1. Repeated blocks

Encrypt

`ABCDEFGH ABCDEFGH ABCDEFGH`

with ECB and CBC.

Compare the ciphertext block patterns.

2. Different IVs

Use the same key and plaintext but

$$IV_1 \neq IV_2.$$

Verify that the CBC ciphertext changes.

3. Error propagation

Flip exactly **one ciphertext bit** and decrypt again.

- Which plaintext blocks change in ECB?
- Which plaintext blocks change in CBC?
- How many bits are modified?

Expected analysis

Explain **why** the mode produces the observed behavior, not only what happened.

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Known-Plaintext Brute-Force Attack

Assume one known plaintext/ciphertext pair:

$$C = E_K(P).$$

For every candidate K':

$$C' = E_{K'}(P).$$

The key is accepted when

$$\boxed{C' = C}.$$

Laboratory restriction

Do **not** search all 2^{56} DES keys. Use a controlled search space.

Controlled unknown bits

Start with

$$n \in \{16, 18, 20\}, \qquad N = 2^n.$$

Increase to 22 or 24 bits only if the machine allows it.

Key construction

Document the mapping:

candidate integer

↓

56 effective key bits

↓

64-bit DES key

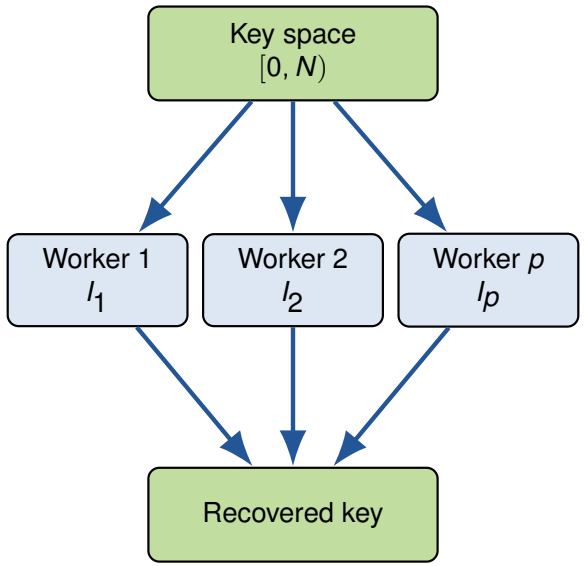
Generate the DES parity bits correctly.

Parallel Exhaustive Search

Partition the search interval among p CPU workers:

$$[0, N) = I_1 \cup I_2 \cup \cdots \cup I_p.$$

- Run experiments with $p = 1, 2, 4, \ldots$
- Each worker searches a different interval.
- Stop the remaining workers when the key is found.
- Record the CPU model and available cores.



Python: for this CPU-bound task, prefer `multiprocessing`.

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Parallel Performance Metrics (cont.)

Throughput

Measures how many candidate keys are tested per unit of time.

$$R_p = \frac{N_{\text{tested}}}{T_p}$$

where:

- N_{tested}: number of tested keys,
- T_p: execution time using p workers.

Unit:

$$\boxed{\text{keys/s}}$$

Example:

$$N = 2{,}000{,}000, \qquad T_4 = 10s$$

$$R_4 = \boxed{200{,}000 \text{ keys/s}}$$

Speedup

Measures how much faster the parallel execution is compared with the sequential execution.

$$S_p = \frac{T_1}{T_p}$$

where:

- T_1: time with 1 worker,
- T_p: time with p workers.

Example:

$$T_1 = 40s, \qquad T_4 = 12s$$

$$S_4 = \frac{40}{12} = \boxed{3.33}$$

The attack is approximately

$$\boxed{3.33\times \text{ faster}}$$

Parallel Efficiency

Measures how effectively the available workers are used.

$$E_p = \frac{S_p}{p}$$

or:

$$E_p(\%) = \frac{S_p}{p} \times 100$$

For:

$$S_4 = 3.33, \qquad p = 4$$

$$E_4 = \frac{3.33}{4} = 0.8325$$

$$\boxed{E_4 = 83.25\%}$$

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Parallel Performance Metrics (cont.)

Performance Metrics

- **Throughput** tells us how many keys are tested per second,
- **Speedup** tells us how much faster the parallel attack is, and
- **Efficiency** tells us how well the CPU resources are being used.

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Performance Metrics and DES-56 Extrapolation

Throughput

$$R_p = \frac{N_{\text{tested}}}{T_p} \quad [\text{keys/s}]$$

Speedup and efficiency

$$S_p = \frac{T_1}{T_p}, \qquad E_p = \frac{S_p}{p}$$

Required plots

Workers vs. keys/s

Workers vs. S_p

Extrapolate; do not execute

Using the best measured throughput R:

$$T_{\max} = \frac{2^{56}}{R}$$

$$T_{\text{avg}} = \frac{2^{55}}{R}$$

Express the result in seconds, days, and years.

Question

Why does adding more workers not necessarily produce perfectly linear speedup?

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Lab 2: Required Work

Part I: Modes of Operation

1. Implement PKCS#7 padding.
2. Implement ECB encryption/decryption.
3. Implement CBC encryption/decryption.
4. Compare ECB and CBC using repeated blocks.
5. Evaluate IV behavior and error propagation.

Part II: Cryptanalysis

1. Implement known-plaintext exhaustive search.
2. Use a controlled unknown-key space.
3. Measure sequential throughput.
4. Parallelize the search across CPU workers.
5. Compute speedup, efficiency, and DES-56 extrapolation.

Implementation rule

The DES core must remain the implementation developed in **Lab 1**; cryptographic libraries may not replace it.

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Lab 2: GitHub and Report

GitHub Repository

The repository must contain:

- DES library from Lab 1.
- ECB, CBC, and padding modules.
- Sequential and parallel brute-force code.
- Automatic tests and benchmarks.
- Experimental results and plots.
- Reproducible `README.md`.

Technical Report

Focus on experimental evidence:

- ECB vs. CBC block behavior.
- IV and error-propagation experiments.
- Keys/s for each worker configuration.
- Speedup and parallel efficiency.
- Estimated exhaustive-search time for DES-56.

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Cryptography

Data Encryption Standard (DES)

Cristhian Iza

School of Mathematical and Computational Sciences
Yachay Tech University

September 2, 2026

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